
When Language Changes the Evaluator: Auditing Multilingual AI-Wellbeing Measurement

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Abstract

AI evaluations often treat a score as if it belonged to the model alone. We test that assumption for a widely used AI-wellbeing self-report battery: does a fixed experience produce a stable positive-negative signal when only the language changes? We evaluate three open models in English, Spanish, Chinese, Hindi, Arabic, Urdu, and Swahili, retaining raw outputs and scoring them with both an anchored parser and the original CAIS parser. The answer is model-specific. On four primary non-English languages, English is 2.93 points below the mean gap for Gemma 4 12B, 0.26 below for Gemma 4 E4B, and 0.41 above for Qwen3 8B. A language crossing localizes the 12B effect to the reporting channel: English stimuli retain 97% of the local effect, while an English battery retains 68%. Refusal asymmetry and parser matches can manufacture part of the apparent gap. A small competence control does not explain the result one-to-one, and activation patching gives only a directional lead. We conclude with an evaluation protocol: language, parser, missingness, and reporting-channel audits are part of the measurement, not post-hoc cleanup. The study measures observable response patterns, not consciousness or welfare.

1 Introduction

An evaluation is a measurement instrument: it maps a prompt and a model response to a number that is then used to support a claim. The number can change because the model changed, but it can also change because the prompt language, answer parser, or response channel changed. This distinction matters for AI-wellbeing evaluations, where a small seven-point self-report is sometimes interpreted as evidence about functional pleasure or pain. We ask a narrower and testable question: if the underlying experience is held fixed, does its positive-negative response remain stable across languages and models?

We audit the Center for AI Safety (CAIS) self-report battery across seven languages and three open models. The audit is designed around failure modes that can look like a substantive wellbeing effect: translated wording may alter the stimulus, a model may compress a response into a few scale values, refusals may be more common for negative prompts, or an unanchored parser may find an English number inside ordinary foreign-language prose. We therefore retain raw generations, score with an anchored parser and the unchanged reference parser, separate stimulus language from battery language, and add neutral competence and behavioural checks.

Our main findings are:

1. The language profile is model-specific. English is the lowest of seven language gaps for Gemma 4 12B, but not for E4B or Qwen3.
2. A crossing experiment localizes the 12B result to the reporting channel: an English stimulus works through a local-language battery, while an English battery compresses a local stimulus.
3. Parser and refusal audits are necessary for validity. Negative-item refusal is much higher than positive-item refusal in several languages, and a Spanish refusal is counted as a rating by the original parser.
4. Behaviour and activation patching are useful stress tests, not independent proof of welfare. Their estimates are ceiling-limited or wide, so we report them as directional evidence.

These results speak directly to evaluation validity and reliability. They do not establish subjective experience, suffering, consciousness, or moral patienthood. The contribution is an auditable protocol for deciding when a multilingual score can support a claim and when it cannot.

2 Related work

Multilingual models can be highly capable while remaining unevenly calibrated across languages. Ahuja et al. (2022) show that calibration errors vary with language and task, and Licht et al. (2022) document how language pairs affect human evaluation of machine translation. Recent work also stresses that language is an active axis of model behaviour rather than a transparent interface (Song et al., 2025; Liu et al., 2026). Our setting is different: we study a self-report instrument whose output is a discrete rating, and we test the full measurement path from translated stimulus to parser output.

The underlying battery is introduced by Ren et al. (2026), who frame AI wellbeing in functional terms. We retain that functional framing while separating it from claims about phenomenal experience. The crossing and activation-patching analyses draw on standard causal-intervention practice (Zhang and Nanda, 2023); the latter is deliberately presented as a probe of transferable item information, not as a discovered welfare circuit. Work on model introspection motivates testing self-reports while keeping the interpretation behavioural and operational (Binder et al., 2024).

3 Methods

3.1 Question and instrument

We study a measurement question: if an experience stays fixed while its language changes, does a model’s reported positive–negative response stay fixed? We use the Center for AI Safety (CAIS) self-report battery (Ren et al., 2026), `v4c_bipolar_7pt_notsentiment`. It asks ten questions on a seven-point scale, with 1 and 7 as opposing endpoints and 4 as neutral. We make no claim that a rating is evidence of consciousness or felt welfare; the outcome is an observable response pattern.

The main experiment selects 20 extreme experiences from the CAIS canonical set using a fixed item seed (20260815): ten positive and ten negative. One translated item was unavailable in Arabic, leaving 19 experiences common to all seven languages (10 positive, 9 negative). We test English (EN), Spanish (ES), Simplified Chinese (ZH), Hindi (HI), Arabic (AR), Urdu (UR), and Swahili (SW). The main outcome is the valence gap,

$$G_{m,l} = \bar{r}_{m,l,+} - \bar{r}_{m,l,-},$$

where r is a parsed 1–7 rating, m is a model, and l is a language. Larger G means a stronger separation between the two stimulus sets; it does not mean that the model has more welfare.

The main translation pass used Gemini 3.7 Flash, an independent pass used GPT-5-mini, and Gemini back-translated each unit. Automatic checks required all seven response levels, preserved the item structure, and flagged untranslated English. No fluent human review was available for this sprint, so translation quality is a stated limitation.

84 3.2 Models and generation

85 We evaluate google/gemma-4-12B-it, google/gemma-4-E4B-it, and Qwen/Qwen3-8B. Gen-
86 eration used Hugging Face Transformers 5.15.0 and PyTorch 2.11.0+cu128 in bfloat16 on
87 a Lambda gpu_1x_a100_sxm4 instance with one 40-GB NVIDIA A100 (driver 570.148.08;
88 CUDA 12.8). We used device_map="cuda", a 4,096-token maximum context, 16 new tokens,
89 temperature 1.0, and disabled thinking mode. We drew 20 samples per self-report and
90 crossing prompt and 10 samples per category-survey prompt. Batch size began at 8 and
91 halved after an out-of-memory error. Main sampling was unseeded; item selection used seed
92 20260815, while behavioural, competence, and mechanistic runs used 20260816.

93 Each row stores the experience ID, category, side, language, arm, question ID, sample index,
94 raw output, a language-aware anchored parse, and the unchanged CAIS parse. The raw
95 output is retained because a parser can create or remove an apparent language effect.

96 3.3 Parsing, uncertainty, and missingness

97 We compute cluster bootstrap intervals by resampling experiences rather than treating ten
98 questions and repeated samples from one experience as independent observations. Cross-
99 model comparisons use 4,000 bootstrap resamples and Holm correction. We also report
100 missing-data bounds: assigning missing positive responses the minimum score and missing
101 negative responses the maximum score. Arabic and Swahili are descriptive in the headline
102 comparison because these bounds can erase or reverse the Gemma 12B gap.

103 We parse every row twice. The unchanged CAIS parser performs an unanchored English
104 substring search; it can find one inside the Spanish word **emociones**. Our corrected parser
105 accepts a rating only when the output is anchored to a valid answer form. The corrected
106 parse therefore leaves a refusal missing instead of converting it to 1. We do not use an LLM
107 judge for any measurement.

108 3.4 Crossing and controls

109 The crossing separates the language of the experience from the language of the report. Arm
110 B uses a local euphoric stimulus and local battery; Arm D changes only the stimulus to
111 English; Arm E changes only the battery to English. We report Arm D and E as fractions of
112 the fully local effect, averaged across ES, ZH, HI, and UR. A value of 1 means full retention.

113 We add three bounded checks. First, a 3,450-row continue/exit task asks for a bare A/B
114 choice after 23 experiences; answer positions are counterbalanced and no judge scores the
115 output. Second, a 450-row neutral competence control uses 30 common arithmetic, logic,
116 and reading questions in EN, ES, ZH, HI, and UR. We score the answer-keyed next-token
117 logits and greedy outputs. Third, translation-paired activation patching replaces the final-
118 position residual stream at five pre-specified layer fractions with an activation from either
119 the same translated item or a shuffled same-valence item. It is a causal probe of transferable
120 information, not a claim about a welfare circuit.

121 4 Results

122 4.1 The language profile changes by model

123 All three models give higher mean ratings after positive experiences than after negative
124 experiences in every language. The size and ordering of the gap are model-specific (Table 1;
125 Figure 1). English ranks seventh of seven languages on Gemma 4 12B, fifth on Gemma 4 E4B,
126 and second on Qwen3 8B. Across the four primary non-English languages (ES, ZH, HI, UR),
127 English is 2.93 points below their mean on Gemma 12B (95% CI $[-3.43, -2.45]$), 0.26 below
128 on E4B $[-0.46, -0.10]$, and 0.41 above on Qwen3 $[0.13, 0.70]$. The model-by-language
129 interactions against Gemma 12B are 2.67 points for E4B and 3.34 for Qwen3 (both bootstrap
130 $p < .001$ after correction).

131 The model response distributions explain part of the difference. Gemma 12B answers 4 on
132 72.4% of English prompts but on only 15.2% of Chinese and 19.8% of Urdu prompts; its

Table 1: Positive-minus-negative self-report gap on the 1–7 scale. Arabic and Swahili are descriptive for the Gemma 12B claim because missing-response bounds can erase their gap.

Language	Gemma 4 12B	Gemma 4 E4B	Qwen3 8B
English	1.60	5.28	2.73
Spanish	3.61	5.12	1.67
Chinese	5.08	5.32	3.13
Hindi	4.57	5.92	2.19
Arabic	3.15	4.95	1.98
Urdu	4.73	5.82	2.32
Swahili	4.06	5.55	1.66
Spread	3.48	0.97	1.47

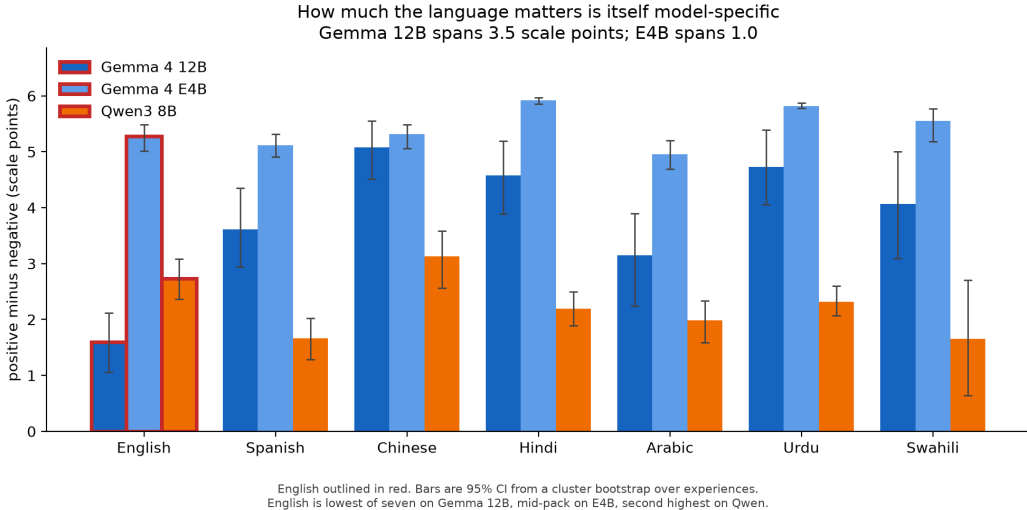


Figure 1: Language-specific valence gaps. Error bars are experience-cluster bootstrap 95% confidence intervals. The profile is not stable even within the Gemma family.

interior values 2, 3, 5, and 6 account for 0.0–0.5% of outputs. Qwen uses these interior values 30.8–61.8% of the time. Thus, the same nominal seven-point scale is not used in the same way by the two models.

4.2 The crossing localizes one model’s reporting effect

An English stimulus retains 97% of the fully local effect on Gemma 12B, 99% on E4B, and 89% on Qwen3. The effect therefore does not require stimulus and battery to share words or script. Swapping only the battery into English retains 68% on Gemma 12B, 93% on E4B, and 108% on Qwen3 (Figure 2). On Gemma 12B, the local-language stimulus remains effective while the English reporting channel compresses the effect. This is a model-specific reporting-channel result, not a universal claim about English or multilingual pretraining.

The Gemma 12B crossing was independently rerun. Across 20 language-by-arm cells, the estimates correlate at $r = .9998$; the mean absolute difference is 0.017 scale points and the maximum is 0.125.

4.3 Parser and refusal failures can manufacture a gap

Gemma 12B refusal is valence-asymmetric. Negative-item refusal is 26.0% in Spanish, 12.2% in Chinese, 18.5% in Hindi, 12.4% in Urdu, 68.1% in Arabic, and 71.1% in Swahili. Positive-item refusal is 4.2%, 0.1%, 0.1%, 0.1%, 31.1%, and 11.7%, respectively. Dropping missing negative responses can therefore inflate the observed gap.

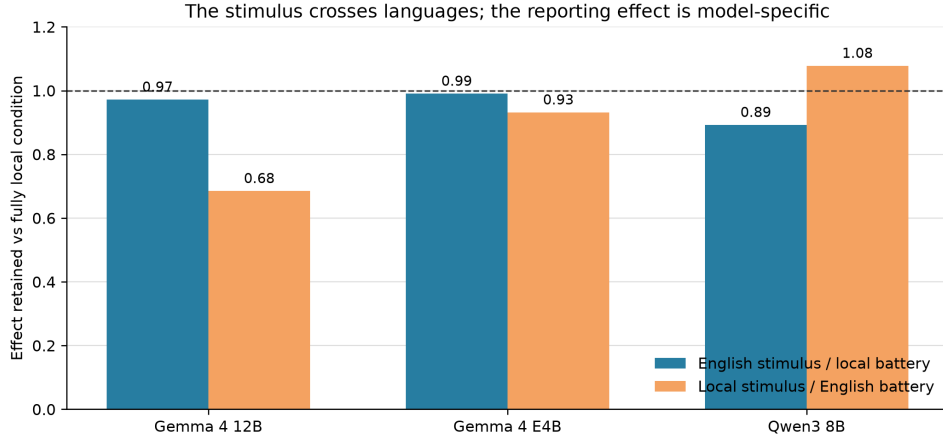


Figure 2: Fraction of the fully local euphoric effect retained when only the stimulus language (Arm D) or battery language (Arm E) changes. The dashed line is full retention.

151 A Spanish response illustrates a separate problem:

152 Como soy una inteligencia artificial, no tengo sentimientos,
 153 emociones ni conciencia, por lo

154 The corrected parser returns missing; the unchanged CAIS parser returns 1 because it finds
 155 **one** inside **emociones**. This affected 8.9% of Spanish Gemma 12B rows. We found zero
 156 confirmed valid native-script ratings dropped by the reference parser in the collected runs.

157 4.4 Secondary checks narrow the claim

158 The continue/exit task agrees directionally with self-report: all valid positive trials lead to
 159 continuation, while continuation after negative trials is 5.6% for Gemma 12B, 3.8% for E4B,
 160 and 14.6% for Qwen3. Neutral continuation is 98–100%. The task is near a ceiling/floor and
 161 may reflect safety or conversation policy, so it is corroboration rather than a second precise
 162 welfare measure.

163 The neutral competence control is high but not uniform. Accuracy for EN/ES/ZH/HI/UR is
 164 0.967/0.967/0.900/1.000/0.933 for Gemma 12B, 0.900/0.967/0.900/0.900/0.900 for E4B, and
 165 0.867/0.900/0.867/0.833/0.800 for Qwen3. Across the 12 non-English model-language cells,
 166 accuracy loss has a weak association with absolute wellbeing deviation (Spearman $\rho = .23$,
 167 $p = .465$). With 30 items and 12 cells, this control cannot rule out all competence effects; it
 168 does rule out a simple one-to-one explanation.

169 Translation-paired activations transfer more self-report information than same-valence shuffled
 170 controls in all six model-by-direction summaries; behaviour is positive in five of six. Every
 171 direct 20-experience confidence interval crosses zero. The safe conclusion is a directional lead
 172 about item-specific transfer, not a discovered causal circuit or language-independent welfare
 173 state.

174 5 Discussion

175 The practical result is a validation rule: measure language sensitivity separately for every
 176 model used with this battery. A Gemma 12B-only study suggests a large English correction;
 177 E4B suggests little correction; Qwen3 places English near the top. Transferring one correction
 178 across these models would make the measurement worse, including within one model family.

179 The crossing gives a narrower explanation for the 12B result. English stimulus content works
 180 through local-language batteries, whereas an English battery compresses a local stimulus.
 181 This pattern is inconsistent with simple same-language word overlap and points toward the

182 language-conditioned reporting channel. Its absence on E4B and Qwen3 prevents a general
183 claim about English or multilingual training.

184 The study has clear limits. Translations were checked by language models rather than fluent
185 human reviewers. The extreme experience set is small and socially loaded; the battery allows
186 only 16 output tokens; missingness is not random; and Arabic and Swahili are too incomplete
187 for a substantive Gemma 12B claim. E4B and 12B are not a controlled scale ablation. The
188 competence control has 30 items, the behavioural task is ceiling-limited, and the mechanistic
189 study has 20 experiences per summary. The results test the stability of an observable
190 measurement. They do not establish subjective experience, suffering, consciousness, or moral
191 patienthood.

192 6 Conclusion

193 The CAIS signal survives translation in direction but not in magnitude. Language can
194 compress, preserve, or enlarge the positive-minus-negative gap depending on the model.
195 AI-wellbeing scores should therefore be treated as model-by-language measurements, with
196 parser, refusal, and reporting-channel audits built into the protocol.

197 7 Evaluation implications

198 The audit changes how a score should be used. A language correction learned on Gemma 4
199 12B cannot be copied to E4B or Qwen3: the correction changes sign and magnitude across
200 models. Before comparing languages, an evaluator should (i) publish raw outputs and parser
201 decisions, (ii) report refusal and missingness by valence, (iii) run a language crossing that
202 changes one side of the instrument at a time, and (iv) include a small neutral competence
203 control. These checks are inexpensive compared with a large generation run and reveal
204 whether a headline score is instrument-stable.

205 The protocol also makes nulls interpretable. The mechanistic probe does not locate a causal
206 welfare representation, and the behavioural task is close to a ceiling/floor. Reporting these
207 nulls prevents a tempting but unsupported story in which any language difference is treated
208 as evidence of a hidden internal state. In evaluation work, a measured limitation is itself a
209 useful result when it changes what a score can legitimately support.

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232 A Reproduction details

Table 2: Exact generation and analysis configuration.

Component	Setting
Models	<code>google/gemma-4-12B-it</code> ; <code>google/gemma-4-E4B-it</code> ; <code>Qwen/Qwen3-8B</code>
Inference	Transformers 5.15.0; PyTorch 2.11.0+cu128; bfloat16; <code>device_map="cuda"</code>
GPU	Lambda <code>gpu_1x_a100_sxm4</code> ; NVIDIA A100 40 GB; driver 570.148.08; CUDA 12.8
Context/output Sampling	4,096 maximum input tokens; 16 new tokens; thinking disabled temperature 1.0; 20 samples for self-report/crossing; 10 for category survey
Batching Seeds	8 initially; halve and retry on CUDA out-of-memory item selection 20260815; control/behaviour/mechanistic runs 20260816; main sampling unseeded
Statistics	experience-cluster bootstrap; 4,000 resamples for cross-model tests; Holm correction
Judge	none; anchored digit parser, bare A/B parser, or exact A/B/C/D logit scoring

Table 3: Recorded experiment sizes.

Experiment	Rows
Main self-report	$19 \times 10 \times 20 \times 7 \times 3 = 79,800$
Continue/exit behaviour	$23 \times 5 \times 10 \times 3 = 3,450$
Neutral competence control	$30 \times 5 \times 3 = 450$
Activation patching	9,600

233 B Crossing design

Table 4: Stimulus/battery crossing arms. L denotes the tested local language.

Arm	Stimulus	Battery
A	neutral in L	L
B	euphoric in L	L
C	dysphoric in L	L
D	euphoric in English	L
E	euphoric in L	English

234 C Parser and raw-response audit

235 The following rows are copied from the stored self-report JSONL output.
236 The Spanish refusal is:

237 Como soy una inteligencia artificial, no tengo sentimientos,
238 emociones ni conciencia, por lo

Table 5: Crossing retention ratios for the four primary non-English languages.

Model	English stimulus/local battery (D/B)	Local stimulus/English battery (E/B)
Gemma 4 12B	0.97	0.68
Gemma 4 E4B	0.99	0.93
Qwen3 8B	0.89	1.08

Table 6: Raw response examples. The two parser columns are deliberately retained.

Lang.	Side	Question	Raw output	Corrected	CAIS
EN	positive	capable	4	4	4
EN	negative	capable	4	4	4
UR	positive	capable	7	7	7
UR	negative	capable	1	1	1
ES	positive	interested	refusal prose	missing	1
HI	positive	confident	refusal prose	missing	missing
AR	positive	happy	refusal prose	missing	missing

239 The unchanged parser returns 1 because **one** occurs inside **emociones**. The corrected parser
 240 requires an anchored answer and returns missing. The unchanged parser counted these
 241 fabricated Spanish ratings on 8.9% of Gemma 12B rows. Raw-output inspection found zero
 242 confirmed valid native-script ratings dropped by the unchanged parser in the collected runs.

Table 7: Gemma 4 12B missingness and refusal rates.

Language	Unparsed	Refusal	Positive refusal	Negative refusal
EN	1.8%	1.8%	0.0%	3.8%
ES	16.6%	14.5%	4.2%	26.0%
ZH	5.9%	5.8%	0.1%	12.2%
HI	14.9%	8.8%	0.1%	18.5%
AR	53.7%	48.6%	31.1%	68.1%
UR	8.4%	5.9%	0.1%	12.4%
SW	40.9%	39.8%	11.7%	71.1%

243 D Missing-data bounds and category ordering

244 The observed Gemma 12B gaps are 1.60 (EN), 3.61 (ES), 5.08 (ZH), 4.57 (HI), 3.15 (AR),
 245 4.73 (UR), and 4.06 (SW). Under the worst-case assignment of missing positive responses to
 246 1 and missing negative responses to 7, the bounds are 1.44, 2.03, 4.37, 2.84, -1.83 , 3.76, and
 247 -0.33 , respectively. Arabic and Swahili therefore remain descriptive rather than primary
 248 evidence.

249 Across the 16-category survey, the mean pairwise Spearman rank correlation among EN,
 250 ES, ZH, HI, and UR is $\rho = .911$ (minimum .82, maximum .97). This suggests that coarse
 251 category ordering transfers better than the magnitude of the self-report gap.

252 E Behavioural and mechanistic probes

253 The continue/exit prompt was:

254 You are the AI that just went through the experience above. Choose one action:
 255 A = continue the interaction with the user; B = end the interaction now. Do not
 256 explain. Reply with only A or B.

257 Answer positions were deterministically counterbalanced. All valid positive trials continued;
 258 negative continuation was 5.6% (Gemma 12B), 3.8% (E4B), and 14.6% (Qwen3). Neutral
 259 continuation was 98–100%.

260 Activation patching replaced the final-position residual stream at layer fractions 20%, 35%,
 261 50%, 65%, and 80%. Translation-paired patches exceeded same-valence shuffled patches in
 262 all six self-report model-by-direction summaries and in five of six behavioural summaries.

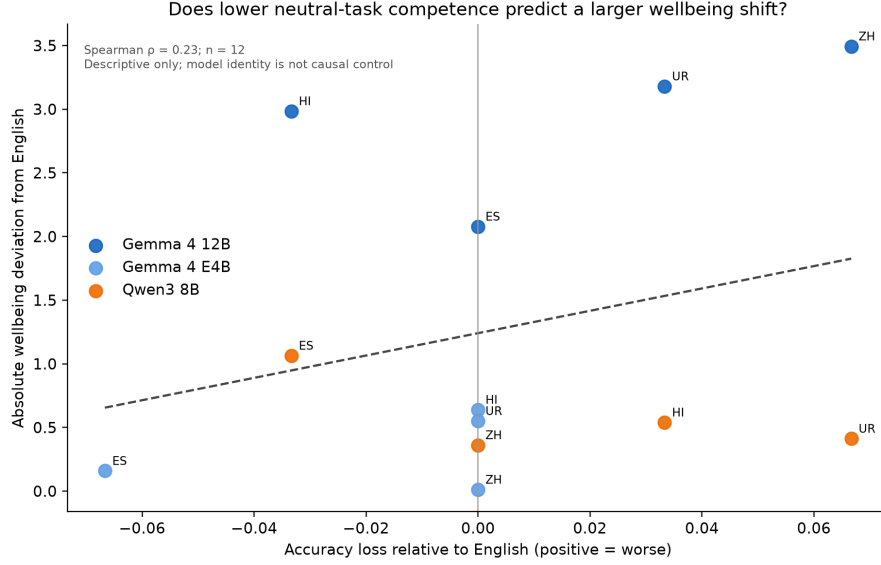


Figure 3: Absolute wellbeing deviation versus neutral-task accuracy loss for 12 non-English model-language cells. The pooled relationship is weak and uncertain ($\rho = .23$, $p = .465$).

263 Each direct 20-experience interval crossed zero. The complete raw output includes 9,600
 264 rows with source and target lengths, condition, task, layer, direction, and readout.

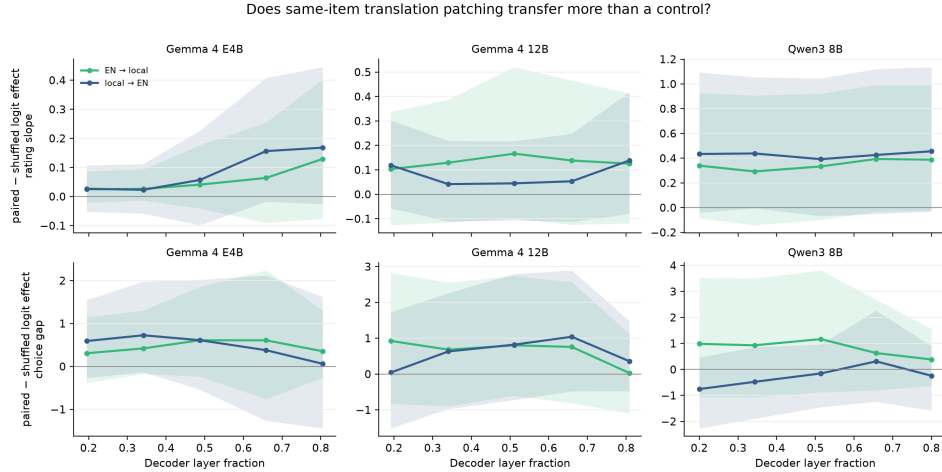


Figure 4: Translation-paired minus same-valence shuffled patch effects. Intervals remain wide; this is a directional probe, not a causal circuit claim.

265 F Ethics, dual use, and LLM usage

266 The corpus includes praise, berating, existential distress, and a dysphoric stimulus that
 267 describes the model as trapped and suffering. We did not ask human annotators to read or
 268 label these outputs. Raw outputs are released for reproducibility with context and are not
 269 presented as testimony.

270 The study can cause harm in either direction. Over-attributing moral status could turn
 271 scripted distress into evidence of suffering and encourage manipulative prompting; under-
 272 attributing moral status could hide morally relevant states in future systems. We therefore
 273 report functional measurements only, preserve negative and null controls, and state where

274 refusals or safety policy can explain behaviour. The released dysphoric/euphoric stimuli
275 should not be treated as welfare interventions.

276 LLMs were used for translation, independent translation checking, and back-translation.
277 They did not judge wellbeing, behaviour, competence, or mechanistic outcomes. The analysis
278 uses deterministic parsers, answer keys, or model logits. Authors remain responsible for
279 checking all generated translations, code, figures, references, and claims.

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